

CHROMOSOME 5 RNA-SEQ ANALYSIS

KAUST Academy Stage 03 - Bioinformatics Project
presented by:

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PRESENTATION AGENDA

Introduction

Project scope, TDP-43 function, and research objectives.

Methodology

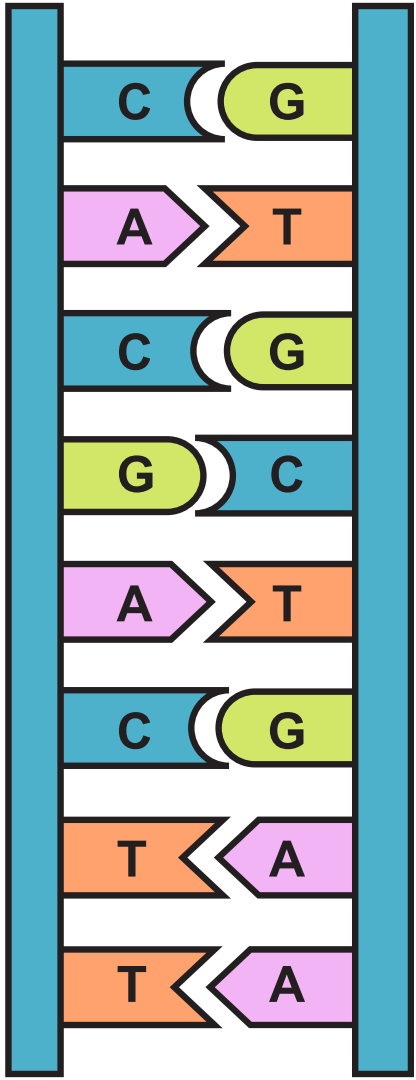
Experimental design, bioinformatics pipeline, and tools.

Data Analysis

Quality control, quantification, and mapping rates.

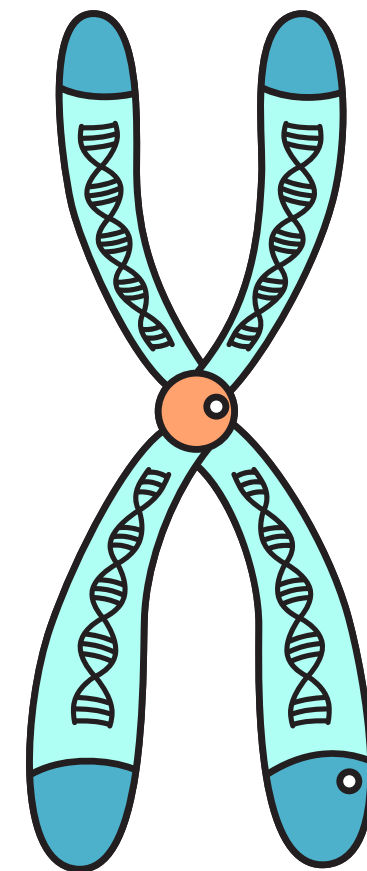
Results

Differential expression, PCA validation, and enrichment.



INTRODUCTION: TDP-43 & RNA METABOLISM

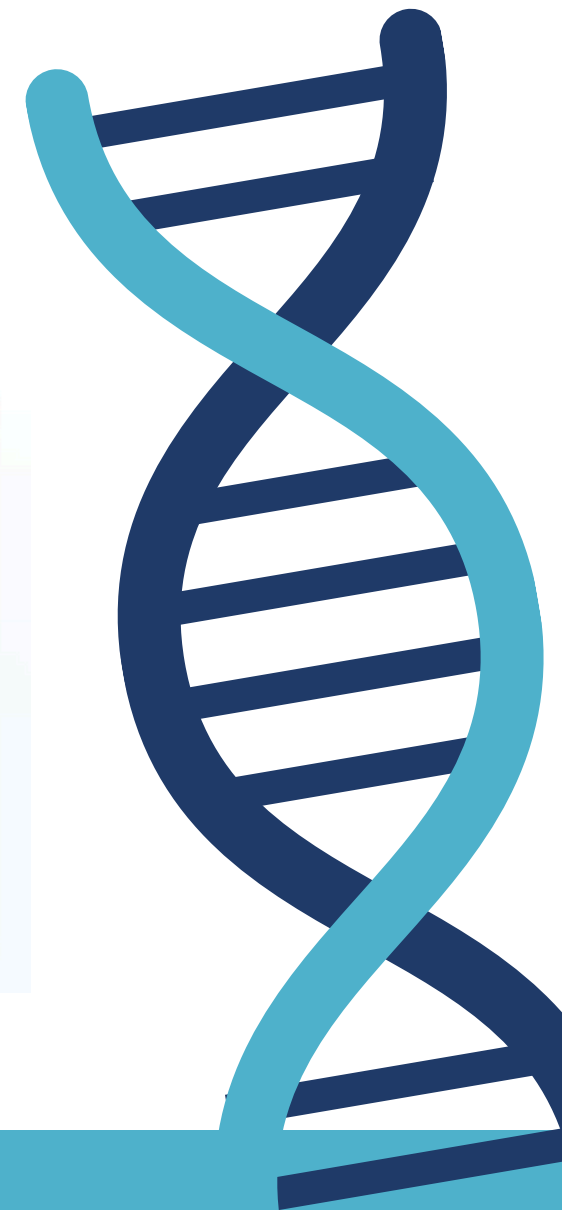
- **Biological Context:** TDP-43 is a critical RNA-binding protein involved in splicing, stability, and transport.
- **Disease Relevance:** Loss of TDP-43 function is strongly linked to neurodegenerative diseases like ALS and FTD.
- **Project Objective:** To analyze the transcriptomic landscape of Chromosome 5 following TDP-43 knockout.
- **Impact:** Identify specific genes and pathways disrupted by the loss of this regulatory protein.



BIOINFORMATICS PIPELINE

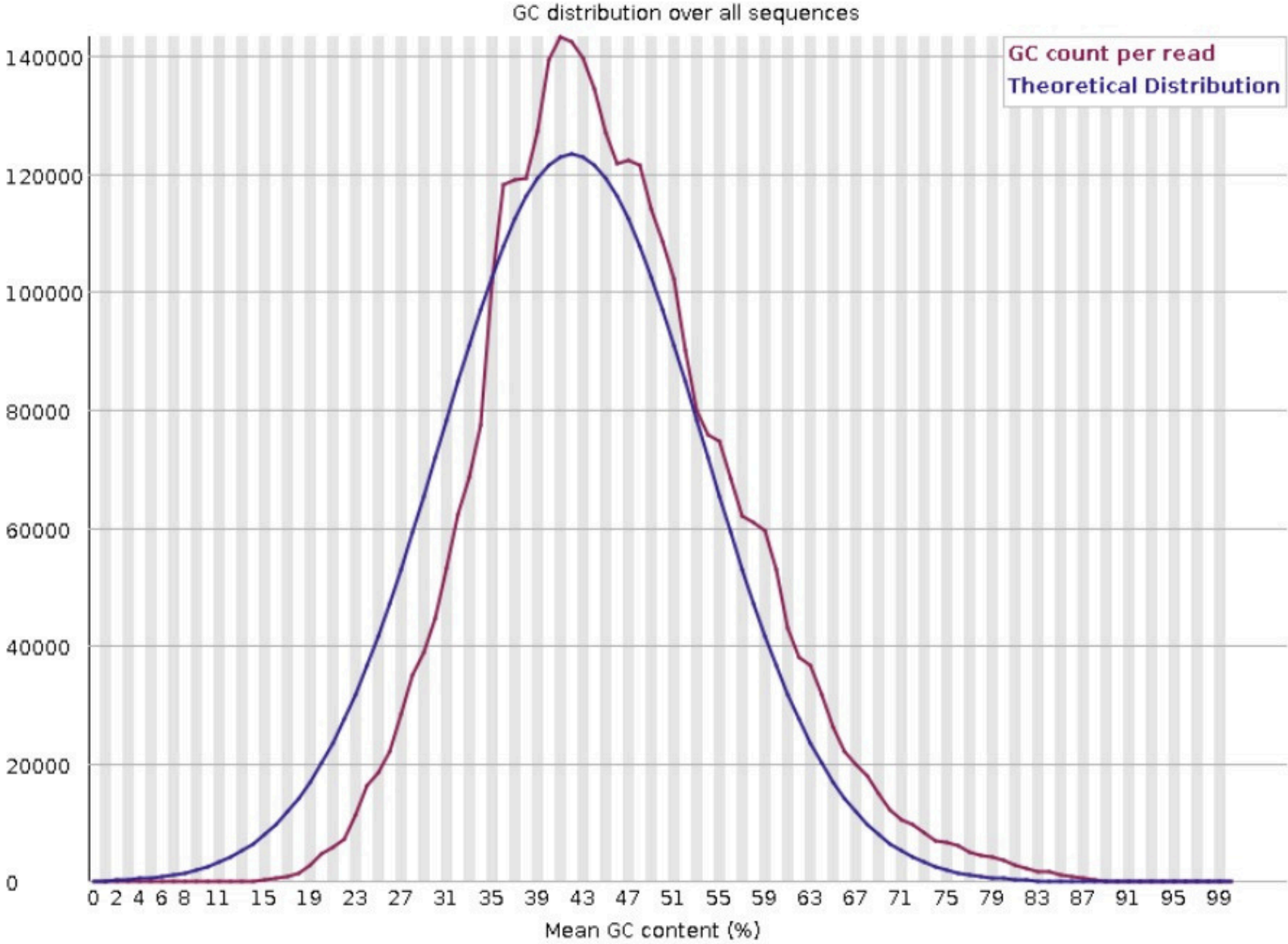


A standardized workflow for reproducible transcriptomic analysis.



CHROMOSOME 5 PROFILE AND DATA QUALIT

Per sequence GC content



FastQC Report

Summary

- ✓ [Basic Statistics](#)
- ✓ [Per base sequence quality](#)
- ✓ [Per sequence quality scores](#)
- ✗ [Per base sequence content](#)
- ! [Per sequence GC content](#)
- ✓ [Per base N content](#)
- ! [Sequence Length Distribution](#)
- ✗ [Sequence Duplication Levels](#)
- ! [Overrepresented sequences](#)
- ✓ [Adapter Content](#)

GC Content Range

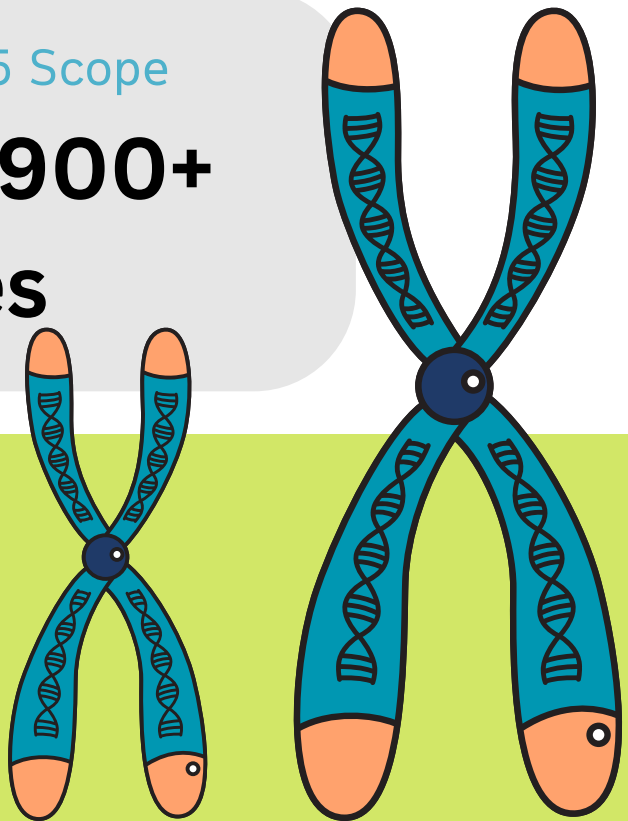
45.0% - 46.2%

Highest Reads (WT_2)

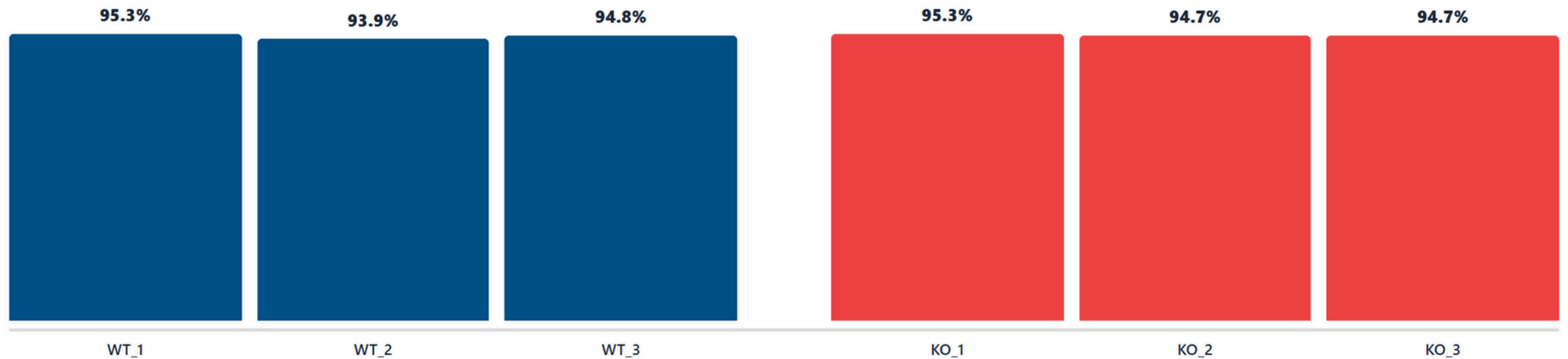
3.3 Million

Chromosome 5 Scope

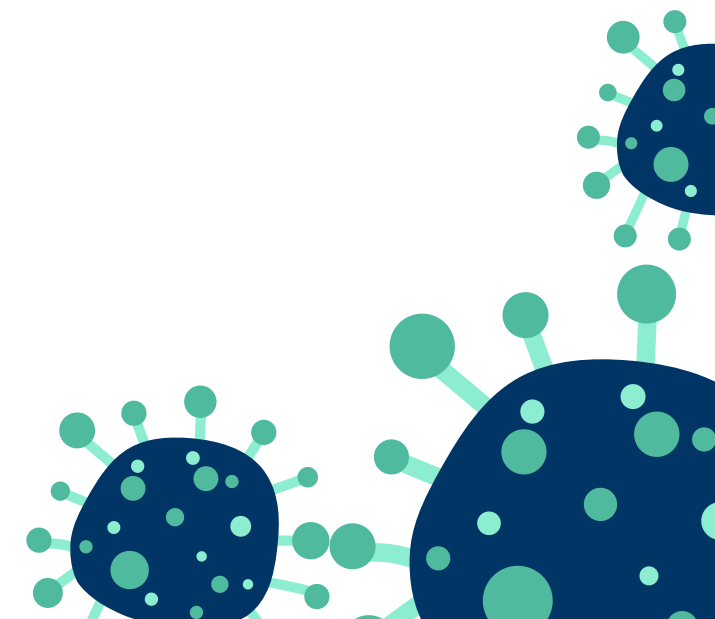
181 Mb / 900+ Genes



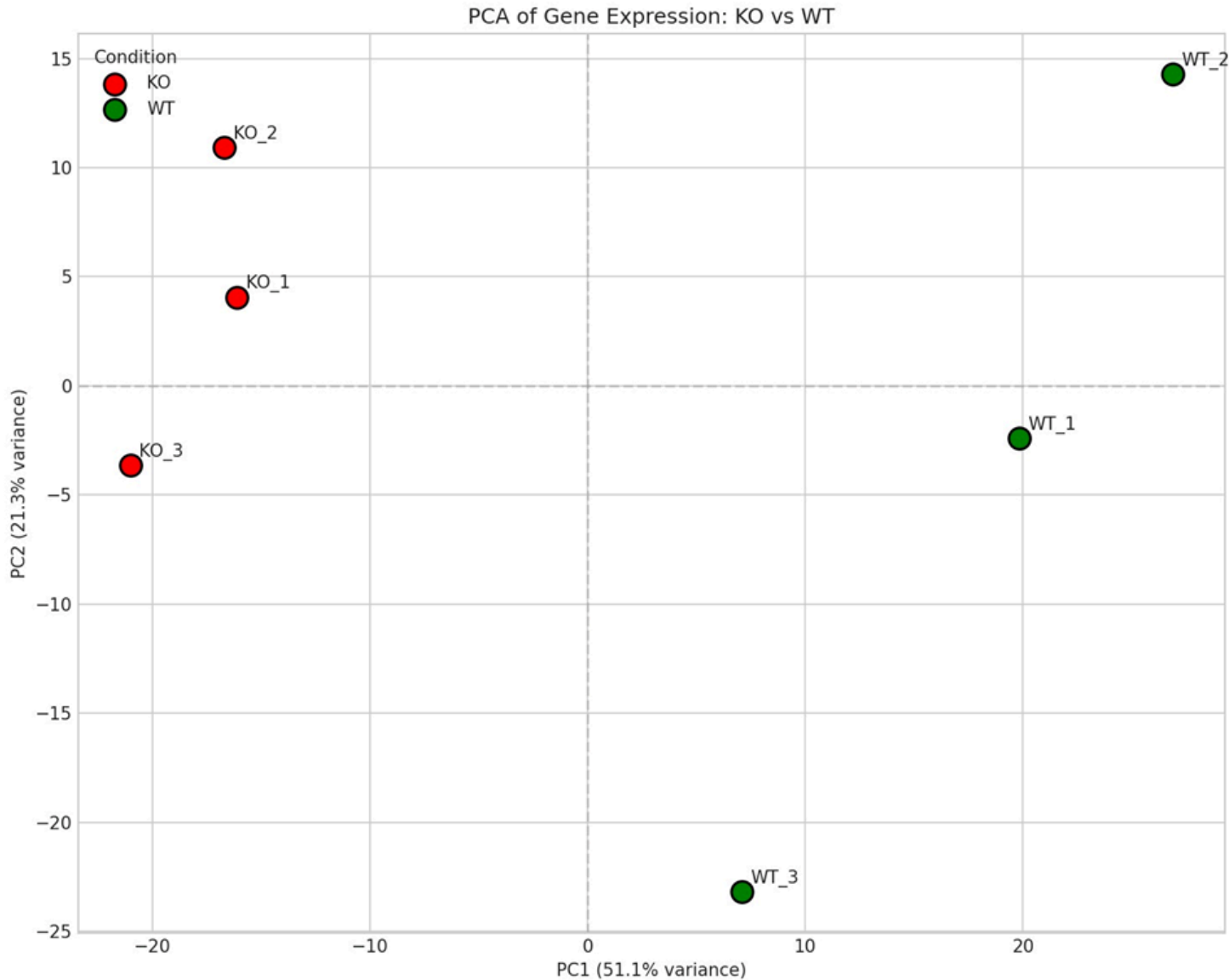
MAPPING & QUANTIFICATION



94.8%
average mapping rate



PCA: BIOLOGICAL VALIDATION



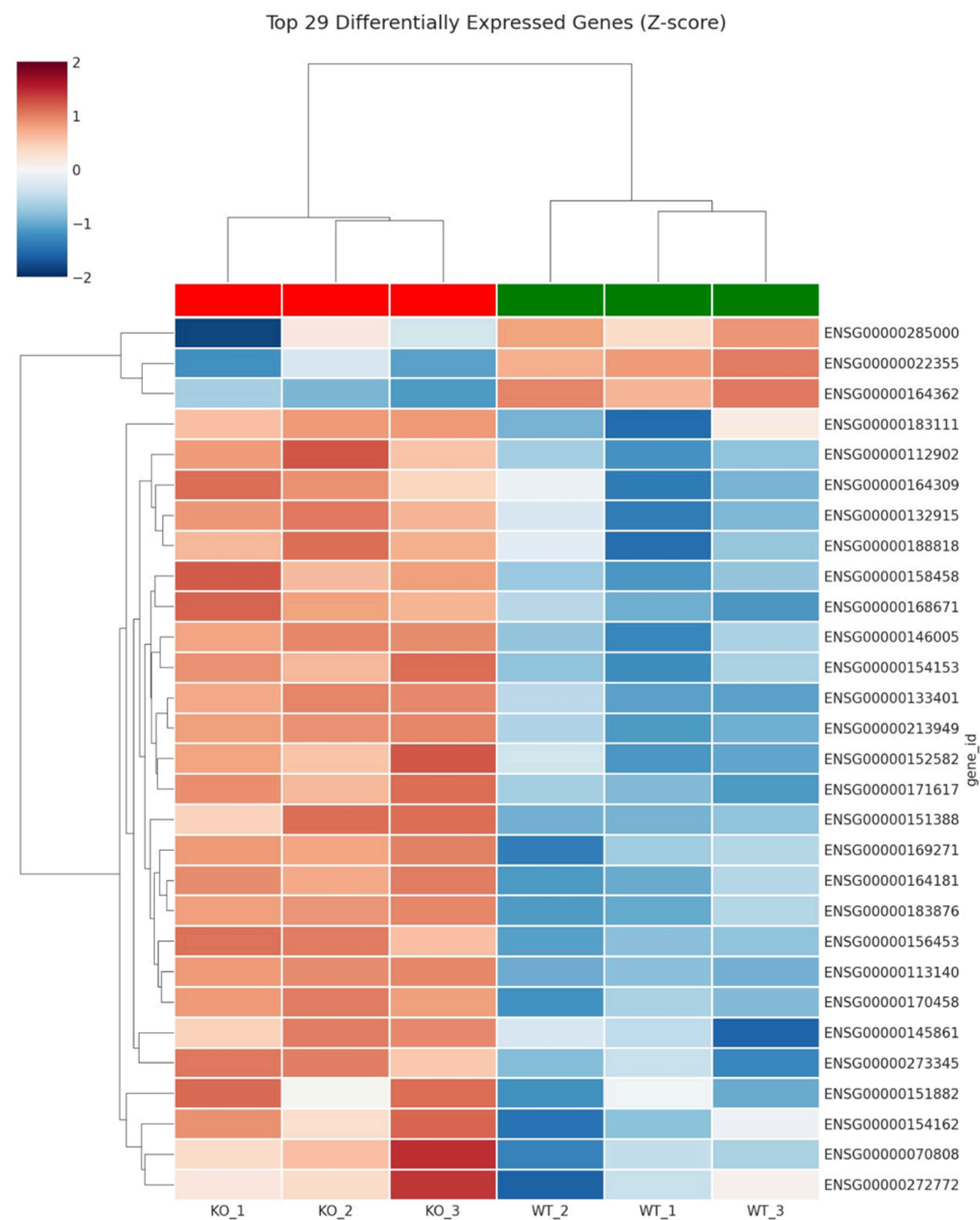
PC1 Variance Explained

51.1%

Observation: Replicates cluster tightly by condition.

Conclusion: PC1 captures the knockdown effect as the primary driver of variance, validating the dataset for differential expression analysis.

DIFFERENTIAL GENE EXPRESSION



Total Significant Genes

29

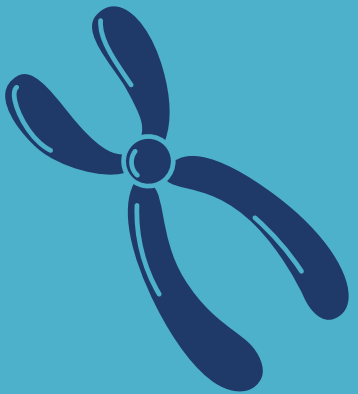
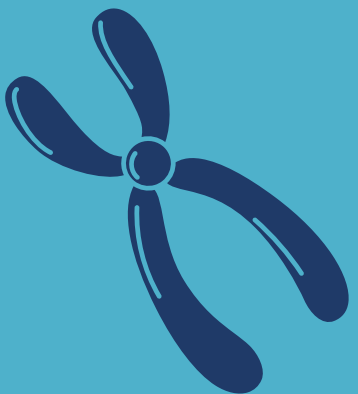
Upregulated Hits

26

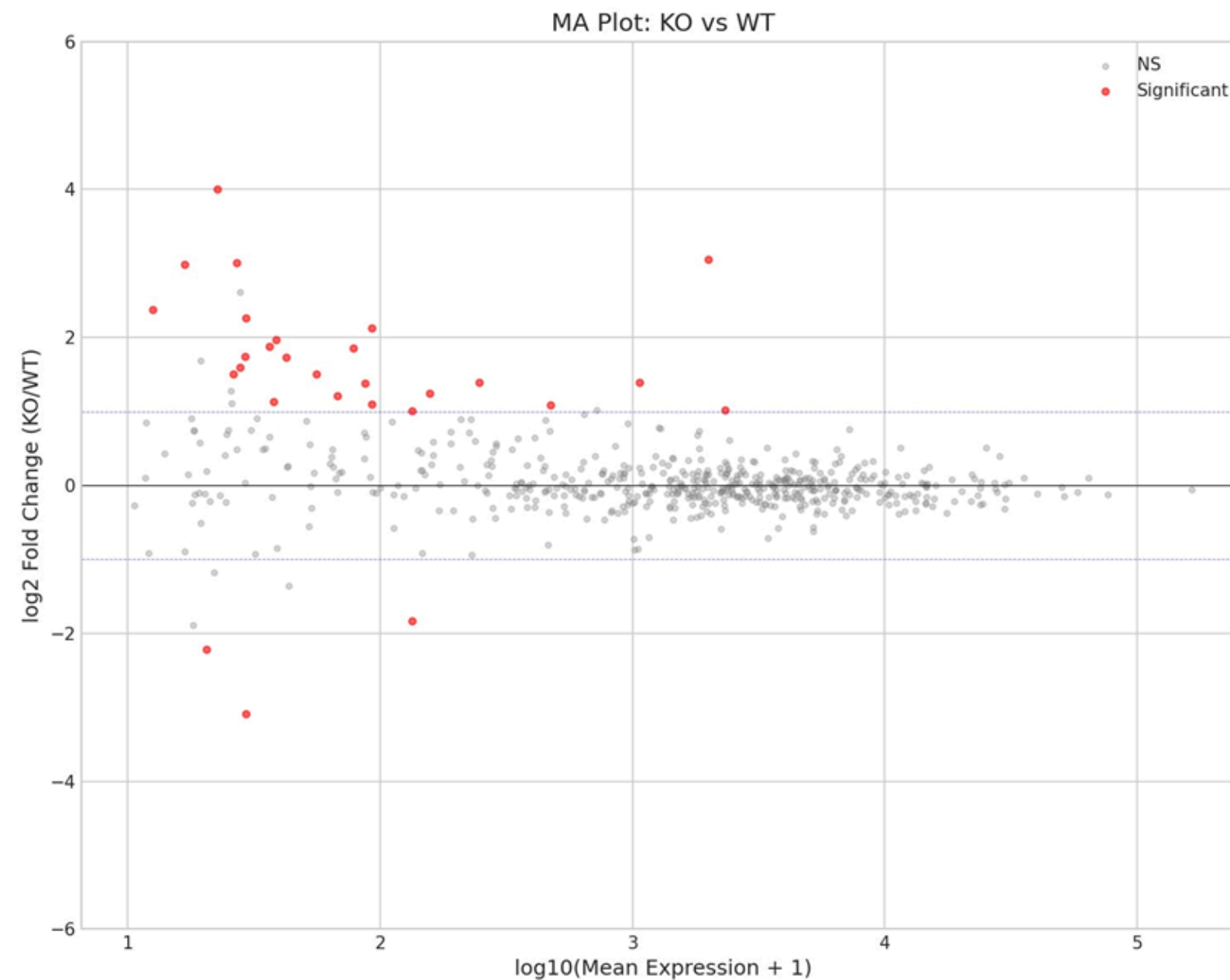
Downregulated Hits

3

Analysis identified 29 significant genes ($p_{adj} < 0.05$), showing a dominant trend of gene upregulation following TDP-43 loss on Chr 5.



FUNCTIONAL ENRICHMENT



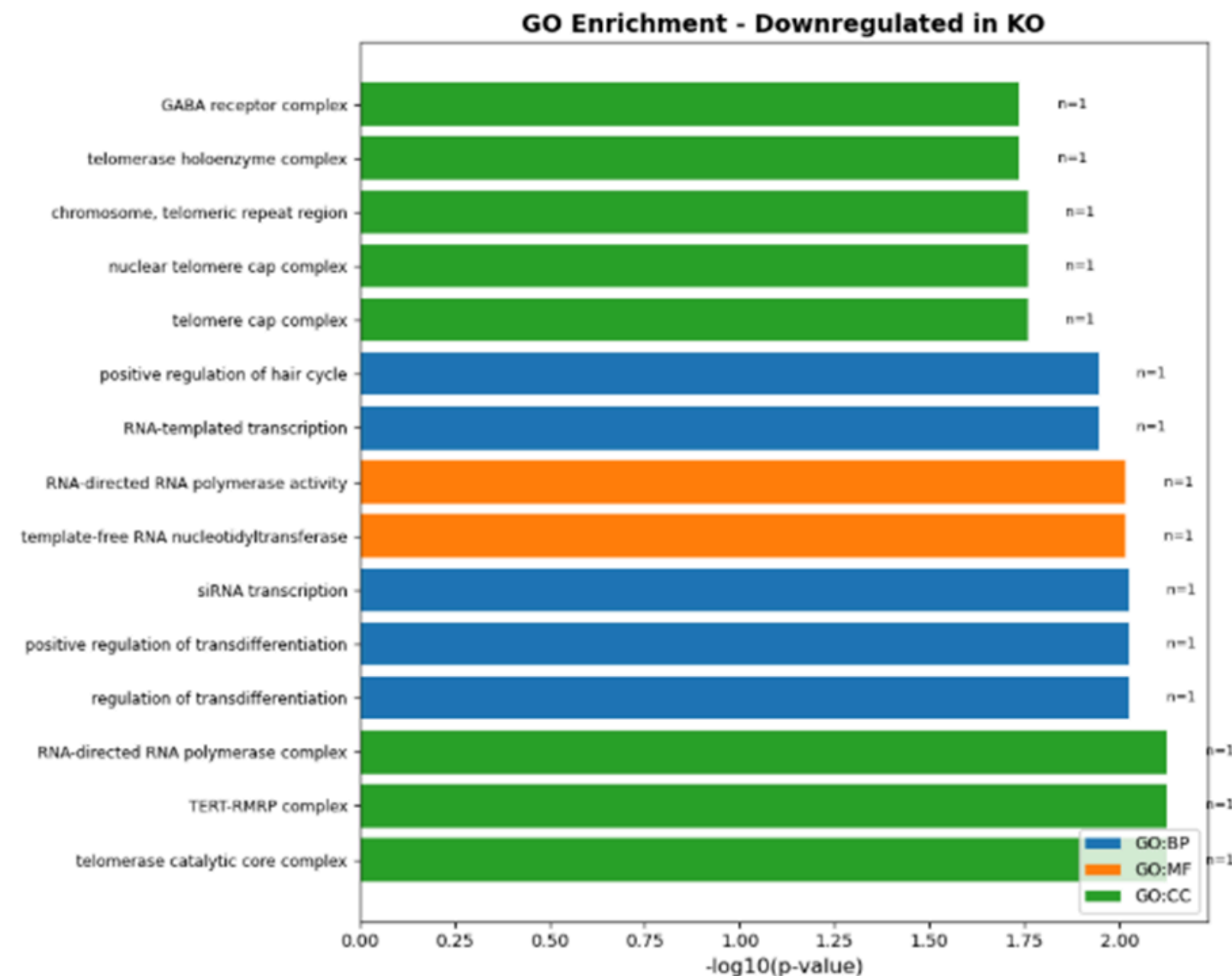
Significant Enriched Terms

150+

Primary disruption occurs in RNA Processing and Transcription Stability. Key enriched terms include the Telomerase catalytic core complex ($p=0.007$).

Biological Link: Confirms localized impact on mRNA metabolism, mimicking ALS/FTD cellular phenotypes.

BIOLOGICAL PATHWAYS AFFECTED



Gene Ontology Analysis

Functional Enrichment

- Primary disruption occurs in RNA Processing and Transcription Stability.
- Key enriched terms include the Telomerase catalytic core complex (p=0.007).

Biological Link:

Confirms localized impact on mRNA metabolism, mimicking ALS/FTD cellular phenotypes.

PROJECT CONCLUSION



Successful Pipeline

Established a robust, chromosome-specific RNA-seq workflow from QC to Differential Expression.



Validated Model

High mapping rates (>94%) and clear PCA separation confirm the integrity of the Knockout model.



Biological Insight

Identified 29 DE genes, revealing that TDP-43 loss primarily drives upregulation and impacts RNA stability.



THANK YOU

Thank you for your attention

